

Answers to Problems 4

1. $Z_{11} = \left. \frac{\dot{V}_1}{\dot{I}_1} \right|_{\dot{I}_2=0} = \frac{1}{Y_1} + \frac{1}{Y_3}, z_{12} = \left. \frac{\dot{V}_1}{\dot{I}_2} \right|_{\dot{I}_1=0} = \frac{1}{Y_2}, z_{21} = \left. \frac{\dot{V}_2}{\dot{I}_1} \right|_{\dot{I}_2=0} = \frac{1}{Y_2}, z_{22} = \left. \frac{\dot{V}_2}{\dot{I}_2} \right|_{\dot{I}_1=0} = \frac{1}{Y_2} + \frac{1}{Y_3}$
2. $y_{11} = \left. \frac{\dot{I}_1}{\dot{V}_1} \right|_{\dot{V}_2=0} = Y_1 + Y_2, y_{12} = \left. \frac{\dot{I}_1}{\dot{V}_2} \right|_{\dot{V}_1=0} = -Y_2, y_{21} = \left. \frac{\dot{I}_2}{\dot{V}_1} \right|_{\dot{V}_2=0} = -Y_2, y_{22} = \left. \frac{\dot{I}_2}{\dot{V}_2} \right|_{\dot{V}_1=0} = Y_2 + Y_3$
3. (a) $A = 1 + \frac{Z_1}{Z_2}, B = Z_1, C = \frac{1}{Z_2}, D = 1$, (b) $A = 1, B = Z_1, C = \frac{1}{Z_2}, D = 1 + \frac{Z_1}{Z_2}, A \leftrightarrow D$
4. $V_1 = 21V, I_1 = 6A$
5.
$$Y = \begin{bmatrix} G_1 + G_3 + \frac{1}{j\omega L} + j\omega C & -G_3 - j\omega C \\ -G_3 - j\omega C & G_2 + G_3 + \frac{1}{j\omega L} + j\omega C \end{bmatrix}$$
6.
$$T = \begin{bmatrix} 1 + \frac{Z_1}{Z_2} & Z_1 + Z_3 + \frac{Z_1 Z_3}{Z_2} \\ \frac{1}{Z_2} & 1 + \frac{Z_3}{Z_2} \end{bmatrix} \begin{bmatrix} 1 + \frac{Z_2'}{Z_3'} & Z_2' \\ \frac{Z_1' + Z_2' + Z_3'}{Z_1' Z_3'} & 1 + \frac{Z_2'}{Z_1'} \end{bmatrix}$$
7.
$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 2/3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$
8. $I_1 = 1A, H_V = 0.25$
9.
$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} 5/3 & 16/3 \\ 1/3 & 5/3 \end{bmatrix} \begin{bmatrix} \dot{V}_2 \\ -\dot{I}_2 \end{bmatrix}, I_1 = 5/4A, I_2 = -5/12A, V_2 = 5/3V$$
- 10.
11.
$$\begin{cases} Y_1 = Y_3 + Y_4 \\ Y_2 = Y_4 \end{cases}, \begin{cases} Y_3 = Y_1 - Y_2 \\ Y_4 = Y_2 \end{cases}$$
12. (1) $V_L = 0.5V$, (2) $V_L = 0.375V$
13. $I_1 = 8A$
14. (1) $R_m = 3\Omega, V_1 = 6V, I_1 = 2A$, (2) $V_{oc} = 4V, V_2 = 2V, I_2 = 2/3A$